current to flow through wires). Another example is gravitational fields (collections of gravitational force lines that, for example, pull us to the Earth's surface).

The Earth's gravitational force lines point radially into the Earth and they pull objects toward the Earth along themselves. The strength of the gravitational pull is proportional to the density of the force lines (the number of lines passing through a fixed area). As they reach inward, the force lines pass through spheres of ever-decreasing area (dotted red spheres in Figure 2.11), so the lines' density must go up inversely with the sphere's area, which means the Earth's gravity grows as you travel toward it, as 1/(the red spheres' area). Since each sphere's area is proportional to the square of its distance r from the Earth's center, the strength of the Earth's gravitational pull grows as $1/r^2$. This is Newton's inverse square law for gravity—an example of the fundamental laws of physics that are Professor Brand's passion in *Interstellar* and our next foundation for *Interstellar*'s science.